

Unit 9, Lesson 4: Finding and Interpreting Median, Mode, and Range

Benchmark

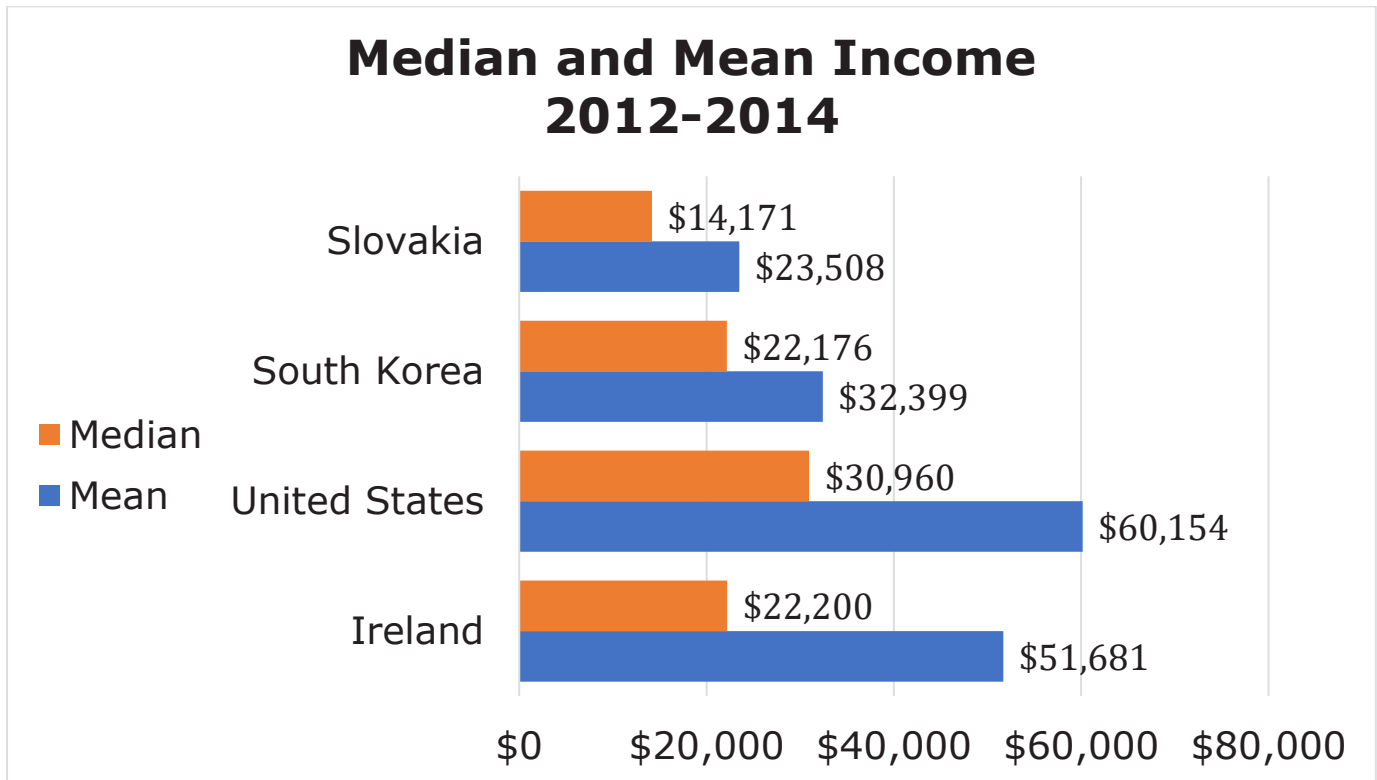
MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

Learning Targets

- ☐ I can find the mean, median, mode, and range of a given data set.
- ☐ I can interpret the median, mode, and range of a given data set.

9.4.1 Warm-Up: Notice and Wonder

The diagram below shows bar charts for the median and mean income between 2012 and 2014 in selected countries. The income is given in U.S. dollars (\$).



1. What do you notice about the values for median and mean income?
2. What do you wonder about the values for median and mean income?

9.4.2 Guided Instruction: Mean, Median and Mode

1. For each definition, fill in the blank with one of the words from the bank.

median

mode

mean

range



The _____ is the arithmetic average of a set of numbers. It is a measure of central tendency.



The _____ is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



The _____ is the value found most often in a set of numbers. There may be no, one, or more than one _____ in the data set. It is a measure of central tendency.



The _____ is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of central tendency.

The number of pets ten students in Tyler’s class have is reflected in the table.

Number of Pets									
1	0	2	1	10	0	2	0	8	12

2. Describe steps to find the median.

3. The median number of pets for the ten students is _____.

4. The mode of the data set is _____.

5. Write an explanation of what the mode of the data set means for the context.

6. Describe the steps needed to find the average number of pets owned by the ten students.

7. The mean number of pets for the ten students is _____.

8. Complete the statement about the mean of the data set.
"Each student in Tyler's class has about _____ pets."

9.4.3 Try It: Mean, Median, and Mode

1. Nine adults boarding a cruise ship at Port Canaveral in Florida were randomly surveyed on the total number of vacations they have taken in the last five years. Their totals are summarized below.

Number of Vacations Taken								
5	4	3	9	1	4	0	6	3

- a. What does the value of the median represent in the context?
- b. What is the median number of vacations taken by the adults surveyed?
- c. Determine the mean of the data. Then, interpret the mean in terms of the context by writing a sentence to summarize. Round to the hundredths.

9.4.4 Exploration: Understanding Mode and Range

Work with your partner to complete the following.

1. The

- ☐ mean
 - ☐ median
 - ☐ mode
 - ☐ range

 is the value occurring most often in

a set of data, which can be used to describe trends in data.

2. While shopping over the last month, Shep collected the price for a jar of pickles from eight stores he visited.

Price for a Jar of Pickles			
Store A	\$1.66	Store E	\$1.99
Store B	\$2.99	Store F	\$1.99
Store C	\$1.49	Store G	\$1.50
Store D	\$1.99	Store H	\$1.99

- a. What does a jar of pickles cost at most stores?

- b. This value is the

- ☐ median of the data set.
 - ☐ mode of the data set.

- c. What is the most Shep will spend on a jar of pickles?

- d. What is the least Shep will spend on a jar of pickles?

- e. The range of a data set describes how spread out the data is. Determine the range in price for a jar of pickles.

3. Shep also began collecting the price of his favorite cereal to determine where he would get the best price.

Price for a Box of Cereal			
Store A	\$3.79	Store E	\$3.99
Store B	\$3.89	Store F	\$3.89
Store C	\$3.99	Store G	\$4.29
Store D	\$3.89	Store H	\$4.59

- a. The mode price of his favorite box of cereal is _____.
- b. What is the most Shep will spend on a box of cereal?
- c. What is the least Shep will spend on a box of cereal?
- d. The range for the price of a box of cereal is _____.
- e. The range describes the

- ☐ center
 - ☐ variability

 of the data set.
4. With your partner, discuss whether the price for pickles or the price for cereal has more variability in pricing. Summarize the discussion.



Measures of variability are numerical values that measure how much a data set varies from a central value.



The range is a **measure of variability**. By comparing the ranges for two or more data sets, the data set that has more variability can be determined by finding which has a larger _____.

9.4.5 Collaboration: Analyzing Numerical Data

Alice collected data on the height of sixth graders in her school. Below are two tables with heights, in inches (in.), of some of her male and female classmates.

Height of Boys (in.)	
58 in.	57 in.
57 in.	56 in.
55 in.	56 in.
57 in.	59 in.
58 in.	57 in.

Height of Girls (in.)	
55 in.	48 in.
55 in.	59 in.
48 in.	50 in.
58 in.	55 in.
50 in.	56 in.

1.

With your partner, determine who will be Partner A and who will be Partner B. Then, follow the directions.

Partner A: Use data from the height of ten boys to find the mean, median, and mode of the boys’ heights in Alice’s class. Then, write each value in the table.

Partner B: Use data from the height of ten girls to find the mean, median, and mode of the girls’ heights in Alice’s class. Then, write each value in the table.
2.

Share the values you calculated with your partner. Then, add the values your partner calculated into your table.

Write a statement in the table interpreting each mean and a statement interpreting each mode.

3. Compare your statements with your partner. If necessary, resolve any differences.

Height of Boys (in.)		Height of Girls (in.)	
Mean of boys' heights		Mean of girls' heights	
Median of boys' heights		Median of girls' heights	
Mode of boys' heights		Mode of girls' heights	
Interpretation of the mean height of boys:		Interpretation of the mean height of girls:	
Interpretation of the mode height of boys:		Interpretation of the mode height of girls:	

4. With your partner, determine the range of boys' heights. Then, determine the range of girls' heights.
5. Explain whether boys or girls in Alice's class have more variability in their heights.

9.4.6 Wrap-Up: Reflection

1. Read each of the “I can” statements below. Check the boxes that best explain your understanding.
 - ☐ I can find and interpret the median, mode, and range of a given data set.
 - ☐ I understand how to find the mean, median, and mode of a data set.
 - ☐ I need support on how to find the mean, median, and mode of a data set.
 - ☐ I understand how to find the range of a data set.
 - ☐ I need support on how to find the range of a data set.
 - ☐ I understand how to interpret measures of central tendency.
 - ☐ I need support on how to interpret measures of central tendency.